

TOUL ROSIERES, France

WMO: 071790

Lat: 48.783N

Lon: 5.983E

Elev: 286

StdP: 97.94

Time Zone: 1.00 (EUC)

Period: 10-19

WBAN: 34051

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-6.2	-4.1	-11.0	1.5	-0.7	-7.9	2.0	-1.3	9.4	9.6	8.4	6.1	1.6	340	0.479

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.0	33.8	20.4	31.2	19.9	29.2	19.2	21.7	30.1	20.9	29.0	20.0	27.8	3.2	130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
19.1	14.4	25.0	18.1	13.5	23.9	17.2	12.7	22.7	64.8	30.3	61.4	29.0	58.3	28.2	27.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.4	6.4	5.6	-8.8	36.9	3.3	2.0	-11.2	38.3	-13.1	39.5	-15.0	40.7	-17.5	42.1		
(5)			-9.2	23.8	3.3	1.6	-11.6	24.9	-13.5	25.9	-15.4	26.8	-17.8	27.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	11.9	4.4	4.4	8.4	10.9	13.4	17.6	19.9	19.5	16.7	13.0	8.1	5.5	(6)
(7)		DBStd	6.62	3.42	4.09	2.96	3.50	3.67	4.32	3.74	3.91	3.42	3.85	3.87	3.49	(7)
(8)		HDD10.0	685	175	159	67	31	9	2	0	0	1	17	81	143	(8)
(9)		HDD18.3	2580	432	391	308	224	158	63	27	32	72	169	306	398	(9)
(10)		CDD10.0	1368	2	2	17	57	116	231	308	295	201	110	26	4	(10)
(11)		CDD18.3	218	0	0	0	1	6	42	77	68	21	3	0	0	(11)
(12)		CDH23.3	3083	0	0	1	37	120	595	1022	955	322	31	1	0	(12)
(13)		CDH26.7	1187	0	0	0	3	24	232	413	422	91	3	0	0	(13)
(14)	Wind	WSAvg	2.3	2.3	2.5	2.6	2.7	2.6	2.3	2.3	2.1	2.0	2.1	2.3	2.2	(14)
(15)	Precipitation	PrecAvg	769	72	60	46	51	65	68	61	59	61	69	80	79	(15)
(16)		PrecMax	966	141	139	111	123	128	141	106	113	135	149	191	211	(16)
(17)		PrecMin	599	12	13	6	8	6	25	18	15	24	32	16	17	(17)
(18)		PrecStd	91	33	38	27	29	27	36	21	30	33	30	42	46	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	14.8	19.1	23.0	27.0	29.2	36.0	37.1	36.0	31.8	26.1	20.8	16.1	(19)
(20)			MCWB	10.8	10.1	11.7	14.8	18.6	20.8	20.4	20.4	19.0	17.3	14.6	11.6	(20)
(21)		2%	DB	12.8	15.1	20.2	24.0	26.9	31.8	33.1	33.8	29.2	23.9	18.1	14.2	(21)
(22)			MCWB	10.2	9.6	11.1	14.0	16.7	21.2	20.5	20.1	17.8	16.7	13.3	10.6	(22)
(23)		5%	DB	11.1	13.0	18.0	21.8	24.0	29.2	30.9	31.2	27.1	22.0	16.1	12.8	(23)
(24)			MCWB	9.4	8.7	10.4	12.4	16.0	19.6	20.1	19.7	17.3	15.9	12.6	10.2	(24)
(25)		10%	DB	9.9	10.9	15.2	19.0	21.9	26.9	28.9	28.9	25.0	20.0	14.2	11.1	(25)
(26)			MCWB	8.4	8.1	9.5	11.4	14.4	18.5	19.3	19.3	16.7	15.2	12.1	9.3	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.8	11.6	13.0	15.8	19.1	24.5	22.7	22.5	20.2	18.6	15.4	12.6	(27)
(28)			MCDWB	13.2	15.9	19.6	24.0	27.6	31.5	31.4	31.2	27.9	23.2	19.1	14.5	(28)
(29)		2%	WB	10.8	10.4	11.8	14.4	17.9	21.6	21.6	21.4	19.3	17.6	14.2	11.7	(29)
(30)			MCDWB	12.3	13.7	17.8	22.5	24.8	29.5	30.0	29.9	26.6	21.6	17.1	13.5	(30)
(31)		5%	WB	9.8	9.4	10.9	13.2	16.7	20.2	20.8	20.7	18.3	16.8	13.4	10.8	(31)
(32)			MCDWB	11.1	12.1	16.2	19.8	22.2	28.0	28.8	28.9	24.6	20.5	15.7	12.6	(32)
(33)		10%	WB	8.6	8.4	10.1	12.2	15.4	19.0	19.8	19.7	17.4	15.8	12.3	9.7	(33)
(34)			MCDWB	9.5	10.7	14.6	17.7	20.4	25.5	27.4	27.4	23.0	19.4	14.6	11.1	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	7.2	9.7	11.3	12.4	12.7	13.4	14.0	14.7	13.4	11.4	8.5	8.6	(35)
(36)			MCDBR	8.7	14.5	17.9	17.1	16.9	17.3	17.5	19.0	17.2	13.9	11.6	11.5	(36)
(37)			MCWBR	6.4	9.2	9.7	8.5	8.4	8.0	7.1	7.2	7.4	7.6	7.3	7.8	(37)
(38)		5% WB	MCDBR	7.5	11.2	13.9	15.3	14.8	15.9	15.7	16.2	14.5	11.4	9.7	8.4	(38)
(39)			MCWBR	6.1	7.9	8.1	8.1	8.2	8.2	7.4	7.4	7.2	6.7	6.5	6.3	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.310	0.334	0.361	0.390	0.400	0.404	0.399	0.396	0.370	0.355	0.326	0.306	(40)	
(41)		taud	2.439	2.386	2.323	2.258	2.267	2.286	2.292	2.328	2.376	2.408	2.437	2.421	(41)	
(42)		Ebn at Noon	755	814	848	856	861	858	856	842	828	776	723	704	(42)	
(43)		Edh at Noon	64	85	106	125	129	128	125	115	100	82	64	58	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.89	1.64	2.86	4.30	5.09	5.75	5.42	4.68	3.58	2.11	1.03	0.71	(44)	
(45)		RadStd	0.14	0.28	0.41	0.56	0.57	0.51	0.52	0.49	0.36	0.17	0.10	0.13	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (46 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page